

CodeGuardian AI - Technical Documentation & User Guide

Enterprise Polyglot Automated Code Inspection & Generative AI Refactoring Engine

1. System Overview & Core Philosophy

CodeGuardian AI combines deterministic static analysis engines (Python AST, Bandit, Radon, Ruff) with Google Gemini 1.5 Flash AI to grade and refactor code across Python, JavaScript, TypeScript, Java, C/C++, Go, Rust, PHP, and SQL.

Key Features:

- **Polyglot Support:** Inspects code across 15+ file formats.
- **Line-by-Line Inspection:** Identifies WHAT is wrong, WHERE it is, and HOW to fix it.
- **1-Click AI Refactoring:** Automatically fixes security bugs, XSS, and SQL injections.
- **Score Monotonicity Guard:** Guarantees that AI refactoring strictly improves quality scores (Score_new >= Score_old).

2. Step-by-Step Running Guide

Step	Action	Command / Details
1	Open Terminal	cd c:\Users\Victus\Desktop\Projects\codeguradian
2	Activate Venv	Windows: venv\Scripts\activate Mac: source venv/bin/activate
3	Migrate DB	python manage.py migrate
4	Run Server	python manage.py runserver 127.0.0.1:8000
5	Open Browser	http://127.0.0.1:8000/ (Login: demo / demo12345)

3. Core Analysis Engine Pipeline

- **ReviewEngine (engine.py):** Orchestrates language routing and enforces Score Monotonicity Guard.
- **AST Analyzer (ast_analyzer.py):** Native AST syntax validation, PEP 257 docstrings, and mutable defaults.
- **Bandit Security (bandit_analyzer.py):** Detects hardcoded secrets (B105), eval() calls (B307), and shell injection (B602).
- **Radon Analyzer (radon_analyzer.py):** Measures Cyclomatic Complexity (Rank A-F) and Maintainability Index.
- **Universal AI Analyzer (universal_analyzer.py):** Multi-language scanner for JS, Java, C++, SQL. Refactors OWASP bugs automatically.
- **Quality Scorer (scoring.py):** Deductive score math (-15 Critical, -10 High, -5 Medium, -2 Low) and Grade classification.

4. Database Schema Overview

- **Project:** Stores owner, name, description, repository_url, and language.

- **CodeReview:** Stores project, version, raw_code, quality_score, letter_grade, summary, and suggested_code.
- **Finding:** Stores review, analyzer, severity, line_number, snippet, title, message, and recommendation.

5. Complete REST API Reference

HTTP Method	API Endpoint Route	Description & Purpose
POST	/api/auth/login/	Authenticates user and returns JWT token
POST	/api/auth/register/	Registers a new user account
POST	/api/projects/	Creates project with optional GitHub URL
POST	/api/reviews/analyze_instant/	Instant live re-analysis endpoint for Code Studio
POST	/api/reviews/fetch_github/	Fetches raw files or file tree from GitHub repository
POST	/api/reviews/audit_repo/	Runs multi-file quality audit across entire repository
GET	/api/reviews/compare/	Compares version histories side-by-side